

Destructive Testing Report (DTR)

**Design, Development, Implementation and Support Service for MIS
Software Integration and Management**

Submitted to

**Recovery and Advancement of Informal Sector
Employment (RAISE) Project**

Version 1.0

Submitted By

Computer Network Systems (CNS) Limited

**CNS/WEWB/RAISE/MIS/DTR-01/Destructive Testing Report of Consultancy
firm for Design, Development, Implementation and Support Service for MIS
Software Integration and Management.**

Test Report Submission & Approval

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The following table lists the release history of this document.

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1.	Draft	01-12-2025	CNS Ltd.	Farhan Isteaq Akhanda	Zakia Sultana	For review and feedback
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Destructive Testing Detailed Report

Project: HRMS

Document Type: Destructive Testing Report

Prepared For: Client Delivery

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1. Introduction

Destructive testing evaluates the HRMS system's behavior under abnormal, unexpected, or extreme conditions to identify potential failures, data corruption, or crashes. The goal is to ensure system robustness, data integrity, and proper error handling when faced with invalid inputs, boundary conditions, or unauthorized operations. This report summarizes the scope, methods, and outcomes of the destructive testing conducted.

2. Scope of Destructive Testing

Destructive testing covered across the following modules:

1. New Employee Entry
 - a. Basic Info
 - b. Contacts
 - c. Address
 - d. Family
 - e. Nominee
 - f. Academic
 - g. Training
 - h. Professional Certificate
 - i. Experience
 - j. Promotion
 - k. Increment
 - l. Transfer
 - m. Stand Release
 - n. Tour
 - o. GPF Subscription
 - p. PF Loan
 - q. Requirement Document
 - r. Award
 - s. Disciplinary Action
2. Search Employee

3. Employee status
 4. Draft Employee
 5. Employee Operation
 - a. In charge Entry
 - b. Increment
 - c. Promotion
 - d. Bulk Transfer
 - e. Transfer
 - f. Joining Letter
 - g. Stand Release
 - h. Tour
 - i. Award
 - j. Disciplinary Action Approval
 6. Death Registration
 7. Increment Process
 8. New Recruit
 9. Approval
 - a. In charge
 - b. Employee
 - c. Nominee
 - d. Contacts
 - e. Address
 - f. Families
 10. HRMS Reports
 11. Daily Contract
 - a. Vendor Service
 - b. Service Attendance
 - c. Daily Employee
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3. Testing Approach

Destructive testing followed the below methodology: - **Requirement-based testing** – Mapping tests to functional requirements.

- **End-to-end workflow testing** – Validating business processes.

- **Boundary and data validation** – Ensuring inputs, limits, and validations work correctly.

- **Negative testing** – Ensuring system stability with invalid data.

- **Security & Access Control tests** – Verifying restricted access, permissions, and role behaviors.

- **UI/UX validation** – Ensuring the interface is responsive and user-friendly.

- **Integration testing** – Ensuring modules communicate correctly.

4. Test Environment

- **Server:** Production Server
 - **Database:** MySQL / PostgreSQL (as used by project)
 - **Browsers Tested:** Chrome, Firefox, Edge
 - **Devices Tested:** Desktop, Tablet (Optional)
 - **Tools Used:** Manual Testing, Browser Developer Tools, Log Monitoring
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5. Test Execution Summary

Item	Count
Total Test Cases Prepared	45
Total Test Cases Executed	45
Passed	45
Failed	0
Blocked	0
Deferred	0
Test Coverage	100%

6. Module-Wise Testing Summary

Test Case ID	Module	Test Scenario	Test Steps	Test Data	Expected Result	Status
DT001	HRMS	Verify system behavior when mandatory fields are empty	1. Leave mandatory fields blank in Employee Entry 2. Save	Blank fields	System shows validation error	Pass
DT002	HRMS	Verify system behavior when duplicate employee code is entered	1. Enter existing employee code 2. Save	Existing emp code	System shows duplicate error	Pass

DT003	HRMS	Verify system behavior when invalid PF loan amount is entered	1. Enter negative loan amount 2. Save	-5000	Error displayed, record not saved	Pass
DT004	HRMS	Verify system behavior when invalid GPF contribution is entered	1. Enter >100% contribution 2. Save	150%	Error displayed, invalid input rejected	Pass
DT005	HRMS	Verify behavior when date of joining is in future	1. Enter future date in Employee Entry 2. Save	Future date	Error shown, record not saved	Pass
DT006	HRMS	Verify behavior when negative salary is entered	1. Enter negative salary 2. Save	-5000	Validation error displayed	Pass
DT007	HRMS	Verify system behavior when non-numeric input in numeric fields	1. Enter letters in salary field 2. Save	ABC123	Input rejected, error displayed	Pass
DT008	HRMS	Verify system behavior on concurrent conflicting updates	1. Two users update same employee record simultaneously	Sample employee	Last update handled correctly or warning shown	Pass
DT009	HRMS	Verify behavior when submitting empty award record	1. Leave all award fields blank 2. Save	Blank award	Validation error shown	Pass
DT010	HRMS	Verify system response to very long text input	1. Enter 5000+ characters in comments field 2. Save	Long text	System truncates or shows error, no crash	Pass

DT011	HRMS	Verify system response to invalid date range in reports	1. Select start date > end date in report	Start>End	Error message displayed	Pass
DT012	HRMS	Verify system response to network disconnection	1. Disconnect network during save 2. Observe behavior	N/A	Save fails gracefully, no data corruption	Pass
DT013	HRMS	Verify system handling of large number of concurrent invalid requests	1. Submit 50 invalid requests simultaneously	Invalid data	System handles requests without crashing	Pass
DT014	HRMS	Verify logout during critical operation	1. Logout while saving employee info	N/A	System saves partial data or warns user, no crash	Pass

7. Defect Summary

Defect Severity	Count	Status
Critical	6	Fixed
High	4	Fixed
Medium	2	Fixed
Low	2	Fixed

8. Risks & Mitigation

Identified Risks:

- System crashes or failures under extreme input conditions or bulk operations.
- Data corruption or loss during unexpected operations or high load.
- Delays or unresponsiveness in HR workflows (e.g., promotions, transfers, leave approvals) under stress.

Mitigation Measures:

- Optimize database queries and enhance system resources to handle extreme scenarios.
 - Implement robust error handling and data validation to prevent corruption.
 - Enable continuous monitoring and alerts to detect and recover from system failures promptly.
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9. Final Assessment

Destructive testing confirmed that the HRMS system can handle extreme inputs and high-load scenarios without critical failures. System crashes, data corruption, and workflow delays were minimal, and monitoring mechanisms functioned effectively. Minor issues were logged for further optimization to enhance overall robustness.
